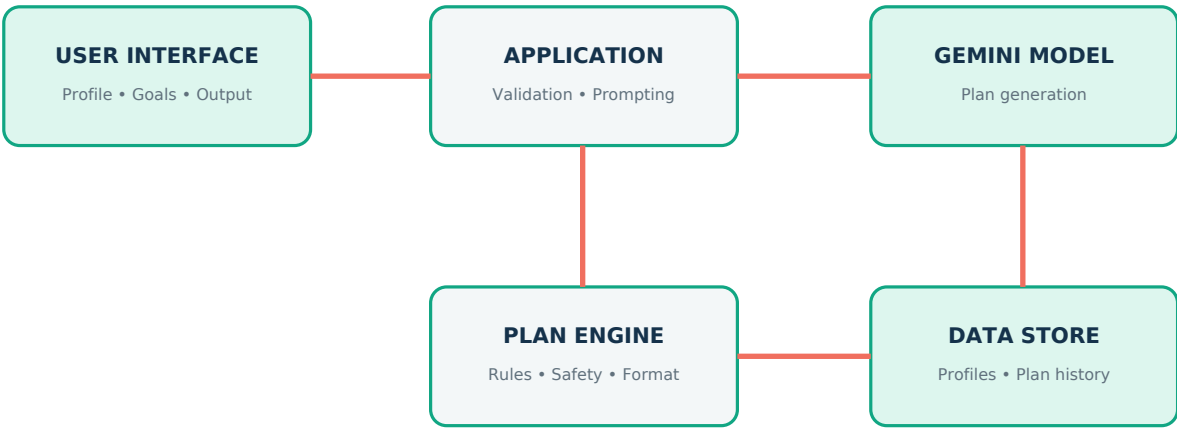


Solution Architecture

Components, responsibilities and safeguards

Date	30 September 2026
Team ID	Not provided
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Team	Lavinraj (Team Leader), Subash, Rahul, Hariharansudhan
Maximum Marks	4 Marks

FitBuddy logical solution architecture



Component	Responsibility	Key safeguard
Profile UI	Collect only inputs required for personalization.	Clear labels, ranges, consent and no medical diagnosis claims.
Request validator	Normalize values and reject invalid combinations.	Allowlisted enums, length limits and server-side validation.
Prompt builder	Convert profile into a controlled model instruction.	Separate system rules from user content; delimit untrusted input.
Gemini adapter	Send request and capture structured response.	Server-only key, timeout and bounded retries.
Plan validator	Check schema, schedule, completeness and prohibited claims.	Reject or repair invalid output before display.
Plan renderer	Present summary, sessions and alternatives.	Escape content; accessible hierarchy; no raw model markup.
History service (optional)	Save plan versions and check-ins.	Authenticated access and least-data retention.

Core generated-plan contract

Object	Required fields
Plan	title, goal summary, weekly frequency, session duration, assumptions, safety note
Day	day label, focus, estimated duration, warm-up, exercises, cool-down
Exercise	name, sets, reps or duration, rest, intensity cue, equipment, easier alternative
Progression	weekly adjustment rule and signs to reduce intensity
Metadata	generation timestamp, model configuration label and plan version